ADAPTIVE PROFILING AND OPTIMIZATION AT THE OPERATOR-LEVEL FOR MACHINE LEARNING MODELS

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BACKGROUND

- **Machine learning (ML)** models are increasingly deployed in real-world environments characterized by: constrained compute, memory, and time
- A lack of real-time profiling and adaptive optimization persists in both **GEMM** and **non-GEMM** workloads. This broader **performance bottleneck** becomes particularly visible in **fake news detection** systems, which often rely on CPU-based models that cannot support real-time inference

MOTIVATION

- Although existing profilers disproportionately focus on **GEMM performance**, non-GEMM operations contribute **11.3–73.6%** of total inference latency in modern **ML workloads** [1]
- This gap motivates our **goal of optimizing these operation-level bottlenecks**, which we demonstrate through a **GPU-accelerated Transformer** for real-time misinformation detection

OBJECTIVE

- Apply profiler to **categorize operations** (i.e. GEMM vs. non GEMM) and **compare** between **CPU and GPU usage**
- Assess how well different **Transformer variants handle fake-news detection** from real world statements using GPUs predominantly

HW & SW SPECIFICATIONS

Software:

- Python 3.10, CUDA 12.6, PyTorch, Torchvision, TensorRT, MATLAB, TensorBoard

Hardware:

- GPU: NVIDIA GeForce RTX 3080 Ti

METHODOLOGY

Models:

- SWIN (Hierarchical Transformer with self-attention windows) [2]
- BERT (baseline vanilla transformer)
- DistilBERT (light weight BERT)
- DistilBERT + TensorRT (real-time optimized inference)
- GPT-2 (generative modeling)

Metrics:

- Accuracy (Correct/Total)
- Inference Execution Time

Datasets:

- COCO 2017 images [3]
- LIAR of short political statements labeled across six truthfulness classes [4]

Steps:

1. Compare SWIN across GPU and CPU w.r.t. to inference time and operation type
2. Run variations of Transformer models on GPU and measure accuracy and inference time
3. Analyze improvements across model types and time

Statement: *"Under the health care law, everybody will have lower rates, better quality care and better access."*
Speaker: Nancy Pelosi
Context: on 'Meet the Press'
Label: False
Justification: Even the study that Pelosi's staff cited as the source of that statement suggested that some people would pay more for health insurance. Analysis at the state level found the same thing. The general understanding of the word "everybody" is every person. The predictions dont back that up. We rule this statement False.

Fig 1. Snippet of LIAR dataset [2]

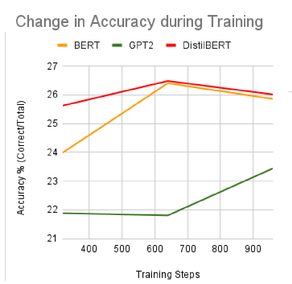

Fig 2. Accuracy at each Validation step across 320 data batches.
DistilBERT + TRT is identical to DistilBERT

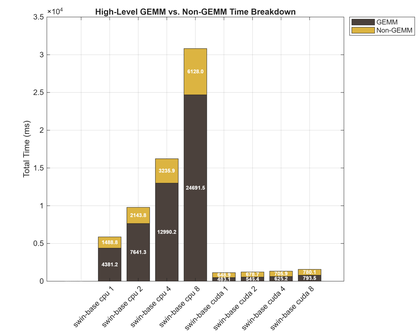

Fig 3. Inference time of operators, CPU vs. GPU for Swin Transformer.

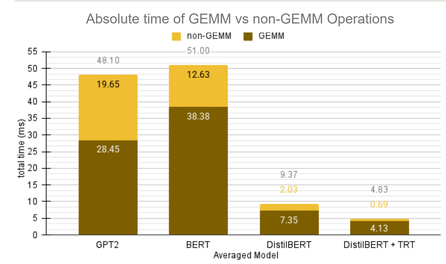

Fig 4. High Level GEMM vs. non-GEMM breakdown across different Transformers

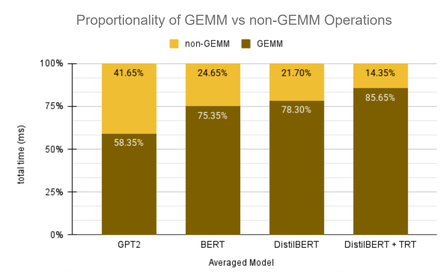

Fig 5. Percentage of time taken by GEMM vs non-GEMM operations across different transformers

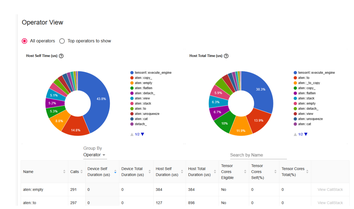

Fig 6. TensorBoard for data visualization

DISCUSSION

- **CPU vs GPU Usage**
  - SWIN execution time was significantly reduced on GPUs compared to CPUs
  - This confirms that GPU-based inference offers superior performance
- **GEMM vs Non-GEMM on GPU**
  - For GPT-2, non-GEMM operations still account for 41.65%
  - For BERT and its variants, no more than 25% of its operations are non-GEMM
  - Suggests that BERT would benefit from a GEMM based optimization
- **Transformers on Real World Dataset**
  - GPT-2 & BERTare transformer models commonly used on text classification
  - All models performed no more than 27% in accuracy across epochs
  - **Time:** DistilBERT+TensorRT performed 544.19% faster than BERT on average
  - **Accuracy:** Model accuracy decreases due to overfitting for BERT and its variants. GPT-2 did not encounter overfitting at this scale

FUTURE WORK

- Propose GPU based method to improve accuracy and decrease execution time
- Explore TensorRT and explain its improvements in execution time in comparison to BERT
- Optimize BERT using different sizes in architecture or using operator level fusion
- Creation of real-time text classification model on tweets or posts

RELATED LITERATURE

[1] R. Karami, S.-C. Kao, and H. Kwon, "Understanding the Performance Horizon of the Latest ML Workloads with NonGEMM Workloads," arXiv.org, 2024. https://arxiv.org/abs/2404.
[2] Liu, Ze, et al. "Swin Transformer: Hierarchical Vision Transformer Using Shifted Windows." ArXiv:2103.14030 [Cs], 25 Mar. 2021, arxiv.org/abs/2103.14030.
[3] https://cocodataset.org/#home
[4] https://www.kaggle.com/datasets/doanquanvietnamca/liar-dataset

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